

## 04 Korean GD / Pan-Asian Graves HLA registry

*Paper 4* 는 *Pan-Asian Graves' disease HLA backlog registry* 다.  
*Source* 조직은 끝났지만 *execution* 은 막혀 있다 – *meta-analysis* 없음, *forest* 없음, *Bundang* 통합 없음, *Paper 2 reactivation* 없음.  
이 페이지는 "*claim boundary* 를 설명하는 *page*" 이지 "지금 분석하는 *page*" 가 아니다.

Status: backlog only

Execution: blocked (4 gates)

Source registry: 15+ indexed

Author: Seungho Cook

Anchor: Paper 2 HT-isolated · Paper 4 GD reserve

Date: 2026-05-04

## I. 한 줄 결론

**핵심 방화벽:** Paper 4 는 GD/Graves disease registry 다. 같은 allele 이름이 Paper 2 와 겹쳐도, 이 페이지의 row 들은 Paper 2 분석 입력이 아니다. Hashimoto-overlap PTC (Paper 2) 와 Graves' disease (Paper 4) 는 disease boundary 가 다르다.

### SOURCES INDEXED

**15+**

Korea, China, Taiwan, Japan, Hong Kong + reviews

### ANALYSES EXECUTED

**0**

No pooled estimate, no forest, no meta

### GATES BEFORE EXECUTION

**4**

Paper 1  
bioRxiv · Paper 2  
closure · Bundang  
n>50 · Yu  
green light

### PAPER 2 BOUNDARY

**FIRM**

GD rows ≠  
Paper 2 inputs

### PROBLEM

East-Asian GD HLA literature 는 풍부 (DPB1\*05:01, B\*46:01, C\*01:02 축) 하지만 source 들 이 산발적이고 harmonization 안 된 상태.

### IDEA

지금은 source 만 망치지 않고 분리해서 모은 다. 나중에 Paper 4 gate 가 열렸을 때 큰 Pan-Asian forest 가능하도록.

### DEFENSE

Paper 2 (HT-overlap PTC) 와 disease boundary 를 명시적으로 분리. shared allele 이름 ≠ shared disease.

### DECISION

Registry yes, execution no. 4 개 gate 통과 후에만 meta/forest 시작.

## 2. 비전공자 배경

**Graves' disease (GD)** 와 **Hashimoto thyroiditis (HT)** 는 둘 다 자가면역 갑상선 질환이지만 임상도 메커니즘도 다르다. GD 는 TSH receptor 자극성 자가항체 + 갑상선 기능항진. HT 는 파괴성 갑상선염 + lymphocytic infiltration. HLA 배경이 부분적으로 겹치는 alleles (DPB1\*05:01, B\*46:01 등) 가 있지만, 한 disease 의 rows 를 다른 disease 의 분석에 그대로 넣으면 결과가 오염된다.

Paper 2 는 Hashimoto-overlap PTC (갑상선암 + Hashimoto 동반) 가 main subject 이고, Paper 4 는 Korean / Pan-Asian Graves disease HLA 가 main subject. **이름이 같아 보이는 allele 이라도 disease 가 다르면 분석 입력이 될 수 없다** 는 것이 이 페이지의 firewall.

## 2.I 용어 정리

| TERM                       | 풀어쓴 설명                                                                   | PAPER 4 에서의 역할                                      |
|----------------------------|--------------------------------------------------------------------------|-----------------------------------------------------|
| GD / Graves' disease       | TSH receptor 자극성 자가항체와 갑상선 기능항진이 중심인 autoimmune thyroid disease.         | Paper 4 의 disease target.                           |
| HT / Hashimoto thyroiditis | 파괴성 갑상선염과 lymphocytic thyroiditis 가 중심인 autoimmune thyroid context.      | Paper 2 context. Paper 4 와 섞지 않는다.                  |
| Pan-Asian registry         | Korea, China, Taiwan, Japan, Hong Kong source 를 한 표로 모으는 planning layer. | 나중에 extraction/meta-analysis 를 준비하는 backlog.        |
| Fine mapping               | allele 뿐 아니라 amino-acid position 까지 disease association 을 좁히는 분석.        | Chu 2018 같은 source 의 강점이지만 여기서는 read-only registry. |
| Bundang integration        | 향후 Korean GD cohort 결합 가능성.                                              | 현재는 금지. n / gate / approval 모두 필요.                  |
| 4-digit HLA                | HLA allele 을 4 자리까지 (예: DRB1*15:01) typing 한 결과.                         | serologic typing (예: DR2) 과 합치면 안 되는 보다 정밀한 layer.  |

### 3. 전체 논리 흐름

Paper 4 는 지금 "더 분석" 하는 단계가 아니라 "나중에 분석할 수 있게 source 를 망치지 않고 분리해 놓는" 단계다. Paper 2 에서 GD evidence 를 섞으면 Hashimoto-overlap PTC story 가 흐려지고, Paper 4 에서 해야 할 Pan-Asian GD HLA story 도 같이 망가진다.

- 1 **GD disease boundary 를 고정한다.** Paper 4 는 GD/Graves 전용이고 HT-overlap PTC 가 아니다.
- 2 **source universe 를 만든다.** Chu, Liao, Shin, Park, Cho, Chen, Ueda, older Japan/Taiwan/HK, reviews/meta 를 registry 로 묶는다.
- 3 **extractability 를 나눈다.** exact counts/OR/p/amino-acid positions 가 보이는 source 와 full table 이 필요한 source 를 분리한다.
- 4 **harmonization 문제를 먼저 적는다.** ancestry, pediatric/adult, GD/GO, serology vs 4-digit, carrier vs allele frequency 를 분리한다.
- 5 **execution gates 를 잠근다.** Paper 1, Paper 2 closure, Bundang  $n > 50$ , Yu green light 전에는 meta-analysis/forest 를 하지 않는다.

#### 3.1 분석 사고 스토리

##### Q1 - 왜 GD 자료를 모으나?

East-Asian GD HLA literature 는 DPB1/B46/C01/A02/DR-DQ 축 이 풍부해서 Paper 4 후보로 강하다.

##### Q2 - 왜 지금 분석하지 않나?

Paper 2 와 Paper 4 boundary 가 먼저 안정되어야 하고, Bundang/ GD execution gate 도 열리지 않았다.

**Q3 - 가장 강한 SOURCE 는?**

Chu 2018 은 Han Chinese GD fine mapping 과 큰 sample size 때문에 top-tier backlog source.

**Q4 - KOREAN BRIDGE 는?**

Shin 2019 는 pediatric AITD/ GD/HD bridge 이고 Park/Cho 는 Korean historical GD baseline.

**Q5 - 가장 큰 위험은?**

HT 와 GD disease mismatch, old serologic typing, ancestry mismatch, frequency metric mismatch.

**FINAL - REGISTRY YES, EXECUTION NO.**

이 페이지는 Paper 4 를 나중에 크게 만들기 위한 지도다. 지금은 pooled statistic 이 없다.

### 3.2 Evidence ladder

| LEVEL | EVIDENCE                                             | READER SHOULD CONCLUDE                                | DO NOT CONCLUDE                              |
|-------|------------------------------------------------------|-------------------------------------------------------|----------------------------------------------|
| 1     | Chu 2018 / Liao 2022 / Shin 2019 / Park 2005 indexed | Core Paper 4 source set exists.                       | It has been harmonized.                      |
| 2     | Country/ancestry map                                 | Pan-Asian scope is plausible.                         | Countries can be pooled without modeling.    |
| 3     | Allele theme map                                     | Candidate allele themes are visible.                  | Allele-level meta-analysis is done.          |
| 4     | Future extraction schema                             | Execution can be planned cleanly.                     | Counts have been extracted source by source. |
| 5     | Paper 2 overlap                                      | Shared allele names can be noted as boundary context. | GD rows validate Paper 2.                    |

## 4. Why this is not Paper 2

**핵심 방화벽:** Paper 4 는 GD/Graves disease registry. 같은 allele 이름이 Paper 2 와 겹쳐도, 이 페이지의 rows 는 Paper 2 분석 입력이 아니다. Paper 1 스타일로 말하면 이 페이지는 "claim boundary 를 설명하는 page" 이지 "지금 분석하는 page" 가 아니다.

### HT VS GD

Hashimoto thyroiditis/HD 와 Graves disease/GD 는 HLA background 가 부분적으로 겹쳐도 임상·메커니즘이 다르다. Paper 2 는 HT-overlap PTC.

### BACKLOG ONLY

Paper 4 는 published GD evidence 를 나중을 위해 organize 한다. Paper 2 의 comparator rows 를 제공하지 않고, 지금 어떤 분석도 trigger 하지 않는다.

## 5. Source registry



| SOURCE                  | COUNTRY / ANCESTRY       | N_GD                    | N_CONTROL                 | HLA LOCI                   | KEY ALLELES / SIGNALS                                 | STATUS           |
|-------------------------|--------------------------|-------------------------|---------------------------|----------------------------|-------------------------------------------------------|------------------|
| Chu 2018                | China / Han Chinese      | 1,468                   | 1,490                     | MHC HLA fine mapping       | DPB1*05:01, B*46:01, DP/DR/B amino-acid positions     | backlog top      |
| Liao 2022               | Taiwanese                | 2,998                   | control design to extract | HLA imputation, class I/II | A*02:07, B*46:01, C*01:02, DP/DQ/DR variants          | backlog high     |
| Shin 2019               | Korean children          | 71                      | 142                       | A/B/C/DRB1/DQB1/DPB1       | B*46:01, C*01:02, DPB1*05:01, DPB1 amino-acid signals | Korean bridge    |
| Park 2005               | Korean                   | 198                     | 200                       | DRB1/DQB1                  | DRB1*08:03, DRB1*16:02, DQB1                          | backlog high     |
| Cho 1987                | Korean                   | 128                     | 220                       | serologic A/B/C/DR         | B13, DR5, DRw8                                        | historical       |
| Chen 2011               | Taiwan / Han Chinese     | table extraction needed | population controls       | A/B/C/DRB1/DQB1/DPB1       | DPB1*05:01, B*46:01, DR/DQ alleles                    | backlog high     |
| Ueda 2014               | Japanese                 | 547                     | 481                       | A/B/C/DRB1/DQB1/DPB1       | B*46:01, DPB1*05:01, DR/DQ alleles                    | backlog high     |
| Older Japan/ Taiwan/ HK | Japan, Taiwan, Hong Kong | varies                  | varies                    | serology / mixed HLA       | Bw46, DR/ DQ signals                                  | source discovery |
| Reviews/ meta           | Asian and mixed          | meta/ review            | meta/ review              | HLA-B, DRB1, broad HLA     | B*46, DRB1, GD/GO review tables                       | citation map     |

## 6. Key allele themes

| THEME                             | PAPER 4 MEANING                                                                 | CURRENT LIMIT                      |
|-----------------------------------|---------------------------------------------------------------------------------|------------------------------------|
| DPB1*05:01                        | Repeated East-Asian GD signal in Chinese/Taiwanese/Japanese/Korean sources.     | No pooled estimate here.           |
| B*46:01 /<br>C*01:02 /<br>A*02:07 | Class I East-Asian HLA block appears across several GD/AITD sources.            | No Paper 2 reuse as GD comparator. |
| DRB1 / DQB1                       | Older Korean GD and newer meta-analysis sources focus on DR/DQ risk/protection. | Resolution varies by study.        |
| Amino-acid fine mapping           | Chu 2018 and Shin 2019 report amino-acid level signals.                         | Registry only; no reanalysis.      |

## 7. What can be done later

### FUTURE EXTRACTION

Published allele/genotype counts, OR/p, amino-acid positions, ancestry labels, phenotype definitions can be extracted source by source.

### FUTURE HARMONIZATION

Allele resolution, serologic vs 4-digit HLA, genotype vs allele frequency, pediatric vs adult GD, GO/comorbidity labels need harmonization.

### FUTURE ANALYSIS

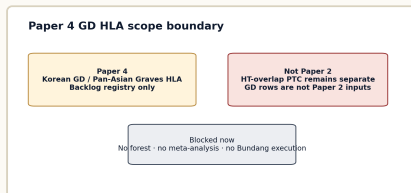
Pan-Asian forest/meta and Bundang integration can happen *only after* Paper 4 gates clear.

### BLOCKED NOW

No meta-analysis, no forest, no Bundang work, no Paper 2 reactivation.

## 8. Figures (P<sub>4</sub>-F<sub>I</sub> ~ F<sub>Io</sub>)

Click any figure to enlarge. These graphics are **registry/status diagrams only**, not pooled-statistic outputs.



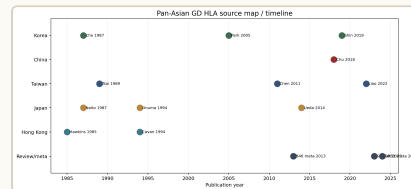
**P4-F1.** What this means: Paper 4 is the GD/Graves backlog. What it does not mean: the backlog is active execution.

## 왜 중요?

Paper 2 와  
Paper 4  
disease  
boundary 를  
시각적으로  
잡는다.

## 분석 사고

좋은 backlog  
는 먼저 무엇  
을 하지 않을  
지 정한다.



**P4-F2.** What this means: sources span Korea, China, Taiwan, Japan, Hong Kong, and reviews. What it does not mean: sources were pooled.

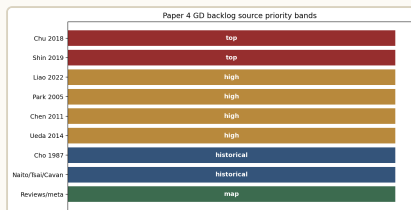
| Paper 4 GD evidence registry matrix (no pooled statistics) |         |         |       |              |              |             |          |
|------------------------------------------------------------|---------|---------|-------|--------------|--------------|-------------|----------|
|                                                            | country | release | ICL   | test         | key studies  | extractable | priority |
| Chen 2010                                                  | China   | 1994    | 1480  | PHX-USA      | DTES & SP202 | yes         | high     |
| Loon 2012                                                  | Norway  | 2008    | 478   | PHX-USA      | ASAC/CP/COG  | yes         | high     |
| Shin 2013                                                  | Korea   | 71      | 242   | ASAC/CP/COG  | PHX-COL/SP2  | yes         | high     |
| Park 2003                                                  | Korea   | 186     | 200   | DAISO        | CHML/COG1    | yes         | high     |
| Chen 2007                                                  | Korea   | 128     | 220   | sewing       | CHML/SP2     | partial     | medium   |
| Chen 2013                                                  | Norway  | 2008    | 478   | ASAC/CP/COG  | DTES & SP2   | yes         | high     |
| Chen 2012                                                  | Japan   | 1987    | 487   | ASAC         | PHX-USA      | yes         | high     |
| Chen 2010b                                                 | Japan   | 1987    | 487   | unidentified | PHX-SP202    | partial     | high     |
| Reichardt et al.                                           | mixed   | 1970s   | 2000s | various      | PHX-USA/SP2  | yes         | high     |

**P4-F3.** What this means: extractability and priority are mapped. What it does not mean: any summary statistic was computed.



Current state: registry ready, execution blocked

**P4-F4.** What this means: Paper 4 requires gates before analysis. What it does not mean: Bundang or meta-analysis work has started.



**P4-F5.** What this means: source priority is staged for future extraction. What it does not mean: priority is an effect size.

#### GD HLA allele theme map (registry only)



**P4-F6.** What this means: common allele themes are mapped. What it does not mean: alleles were pooled across countries.

#### Paper 4 execution block dashboard



**P4-F7.** What this means: only registry work is active. What it does not mean: meta-analysis is allowed now.

#### Future extraction schema (not executed)



**P4-F8.** What this means: future extraction columns are defined. What it does not mean: extraction or harmonization has been completed.

#### Paper 2 / Paper 4 separation wall



**P4-F9.** What this means: the disease boundary is explicit. What it does not mean: shared alleles make GD a Paper 2 comparator.

#### Paper 4 future work ladder



**P4-F10.** What this means: only the registry rung is active now. What it does not mean: later rungs can start without gates.

## 9. Expanded registry planning tables

### 9.1 Future extraction schema

| COLUMN         | WHY IT MATTERS                                                                       | CURRENT PAGE STATUS |
|----------------|--------------------------------------------------------------------------------------|---------------------|
| source_id      | Prevents Chu/Liao/Shin/Park rows from being mixed silently.                          | planned             |
| disease_label  | GD, GO, HD, AITD, control must stay separate.                                        | planned             |
| ancestry       | Korean, Han Chinese, Taiwanese, Japanese, Hong Kong Chinese are not interchangeable. | planned             |
| frequency_type | Allele, carrier, genotype, and serology frequencies cannot be merged silently.       | planned             |
| effect_stat    | OR/p values stay source-level until Paper 4 gates clear.                             | not pooled          |

### 9.2 Blocked-vs-future table

| TASK                 | NOW                                    | LATER PAPER 4                    |
|----------------------|----------------------------------------|----------------------------------|
| GD source registry   | allowed and complete enough for review | expand with full-text extraction |
| Allele harmonization | rules only                             | execute after gates              |
| Forest/meta-analysis | blocked                                | possible after Paper 4 approval  |
| Bundang integration  | blocked                                | possible if n>50 and Yu approves |
| Paper 2 reuse        | blocked                                | still no; Paper 2 remains HT/PTC |

## 10. Safe wording vs unsafe wording

| TOPIC     | SAFE WORDING                                                        | UNSAFE WORDING                                                        | REASON                                            |
|-----------|---------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------|
| Registry  | Paper 4 has a curated source registry for future GD HLA extraction. | Paper 4 has completed meta-analysis.                                  | No pooled statistics were run.                    |
| Chu 2018  | Chu 2018 is a top-priority Han Chinese GD fine-mapping source.      | Chu 2018 is a Paper 2 comparator.                                     | GD rows are blocked from Paper 2.                 |
| Pan-Asian | Asian source coverage is broad enough to plan harmonization.        | Korean, Han Chinese, Taiwanese, Japanese sources are interchangeable. | Ancestry and typing differences require modeling. |
| Bundang   | Bundang integration is a future gate-dependent step.                | Bundang is now active.                                                | User boundary keeps Bundang GD work blocked.      |

## II. Claim boundary table

| CLAIM               | STATUS           |
|---------------------|------------------|
| GD source registry  | YES              |
| Meta-analysis       | NO               |
| Forest plot         | NO               |
| Bundang integration | NO               |
| Paper 2 input       | NO               |
| Future Paper 4      | YES, after gates |

Next action: keep registry frozen, or return to Paper 1 Hook.

## I2. 데이터 / 다운로드

### 📁 PAPER4-HLA/

Paper 4 GD HLA  
detailed assets

### ← PAPER 2 HLA SOURCE

HT-overlap PTC  
source page

### ← PAPER 4 MAIN

Korean GD HLA  
detail

### ← HUB INDEX

8-papers portfolio

---

**Status.** Paper 4 = backlog only. Per memory paper\_numbering\_2026\_05\_04 + v19\_paper4\_GD\_backlog: Paper 4 is gated 4/4 and remains reserve until Paper 1 bioRxiv + Paper 2 closure + Bundang n>50 + Yu green light.

**Live URL.** [http://40.82.129.113:8012/papers\\_hub\\_2026\\_05\\_04/  
paper4\\_gd\\_hla.html](http://40.82.129.113:8012/papers_hub_2026_05_04/paper4_gd_hla.html)